

ENEL700 Tutorial 9

1. An information source is characterized by the following alphabet and probability distribution

Alphabet	A1	A2	A3	A4	A5
Probability	0.1	0.4	0.3	0.15	0.05

Compute the Entropy for the source.

2. Suppose the following code is used for the information source described in question 2, determine its coding efficiency.

Alphabet	A1	A2	A3	A4	A5
Code	01	0	1	00	001

3. The American Standard Code for Information Interchange (ASCII) has 128 characters, which are binary coded. If a certain computer generated 100,000 characters per second, determine the minimum bandwidth required to transmit this signal (Hint: assume a noiseless channel).
4. A television signal has bandwidth of 4.5MHz, and it is sampled, quantized and binary coded to obtain a PCM signal. Suppose the sampling rate is a rate 20% above the Nyquist rate and the number of quantization levels are 1024. Determine the minimum bandwidth required to transmit this signal.